

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. Branch : Electronics and Telecommunication Engineering

Semester : VIII

Max Marks: 60

Subject Code & Name: Cryptography and Network Security(BTETPE802E)

Date: 07/07/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All Questions are Compulsory
2. Draw neat diagram wherever necessary.
3. Figures to right indicates full marks
4. Assume suitable data wherever necessary and mention it clearly

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Write a detailed note on Digital signatures		06
B) Describe about Hash Function. How its algorithm is designed? Explain its features & properties		06
C) Explain Plaintext , Secret key, Cipher text, Decryption algorithm components of encryption algorithm. (Any two)		06
Q.2 Solve Any Two of the following.		
A) Illustrate about the SHA algorithm and explain		06
B) Explain public key and private key cryptosystem		06
C) Explain RSA Approach, DSS Approach two approaches of Digital Signature		06
Q. 3 Solve Any Two of the following.		
A) Explain Substitute bytes, Shift Rows, Mix Columns and Add Round Key operations used in AES (Any two)		06
B) Explain the RSA algorithm and explain the RSA with $p=7, q=11, e=17, M=8$. Discuss its merit		06
C) Explain in detail about elliptic curve cryptography		06
Q.4 Solve Any Two of the following.		
A) Explain about AES in detail		06
B) Let P, C, K denote plaintext space , Cipher text and Key space respectively. In shift cipher $C=P+K \pmod{26}$ suppose the key for shift cipher is $K=6$ and Cipher text is 1 then what is Plain text.		06
C) Specify and explain in details the four categories of security threats		06
Q. 5 Solve Any Two of the following.		
A) Explain Any two out of Electronic Code Book (ECB), Cipher Block Chaining (CBC), Cipher Feedback (CFB), Output Feedback (OFB), Counter Mode different modes of operation in DES		06
B) Explain Classical cryptosystems and its types.		06
C) Define Euler's theorem and its application also Find $\gcd(24140, 16762)$, $\gcd(1970, 1066)$ using Euclid's algorithm		06

***** End *****